

When direct sunlight reaches Golden

Terrain-corrected sun times for the townsite, and how they were produced

Prepared for Mayor and Council, Town of Golden | 6 September 2026

Joy Guyot, Councillor | joy@luminaryx.ca

Code and data: github.com/LuminarAiConsultancy/golden-sunlight

This is personal work by a councillor. It is not a Town of Golden publication.

1. Why this document exists

Every published sunrise table, almanac and news figure used in the time observance discussion assumes a flat, unobstructed horizon. Golden does not have one. The town sits on the floor of the Columbia Valley with the Purcells on one side and the Rockies on the other, so the sun is already well up before it clears the ridge, and it disappears well before it sets.

This document sets out what the difference actually is, in minutes and in days per year, under both clock options. It also sets out how the numbers were produced, where the elevation data came from, how the calculation was checked, and what it does not establish. The intent is that any member of Council or staff can either follow the method on paper or run the code themselves and obtain the same figures.

A note on names, because the terminology is confusing

British Columbia's new permanent "Pacific Time" is UTC-7 all year, which is the same clock reading as Mountain Standard Time. In this document, "Standard time" and "UTC-7" therefore describe the option Council adopted on 1 September 2026. "Daylight time" and "UTC-6" describe permanent Mountain Daylight Time, the option adopted by the Regional District of East Kootenay on 14 August 2026.

The mountains delay sunrise by the height of the skyline

Schematic, not to scale. Winter solstice, standard time. The 42 minute gap is the mountains, not the clock.

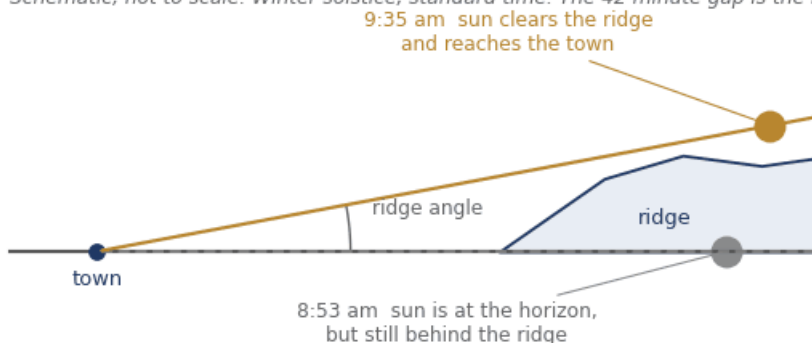


Figure 1. Why a valley town sees the sun later than the almanac says.

2. What the calculation found

2.1 The winter solstice

21 December 2026, at the townsite. Flat-horizon rows are produced by the same program with the terrain switched off, so the comparison isolates the effect of the mountains and nothing else.

	Standard time (UTC-7)	Daylight time (UTC-6)
Sun clears the ridge	9:35 am	10:35 am
Sun leaves the town	3:26 pm	4:26 pm
Direct sun on the town	5 h 52 m	5 h 52 m
Flat horizon would give sunrise at	8:53 am	9:53 am
Flat horizon would give sunset at	4:39 pm	5:39 pm
Flat horizon day length	7 h 47 m	7 h 47 m

Two things follow from that table. The mountains remove 42 minutes from the morning and 73 minutes from the afternoon, and they do so under either clock. The clock does not change how much sunlight Golden receives; it changes which end of the day that sunlight lands on.

2.2 The year in days

Counting every day of calendar year 2027 on which the sun has not yet cleared the ridge at a given clock time:

Sun still behind the ridge at	Standard time	Daylight time
8:00 am	166 days	201 days
9:00 am	104 days	166 days
10:00 am	0 days	104 days

The bottom row is the one that is hardest to obtain from a published table. Under standard time there is no day in the year when the sun has failed to reach the town by ten in the morning. Under daylight time there are 104.

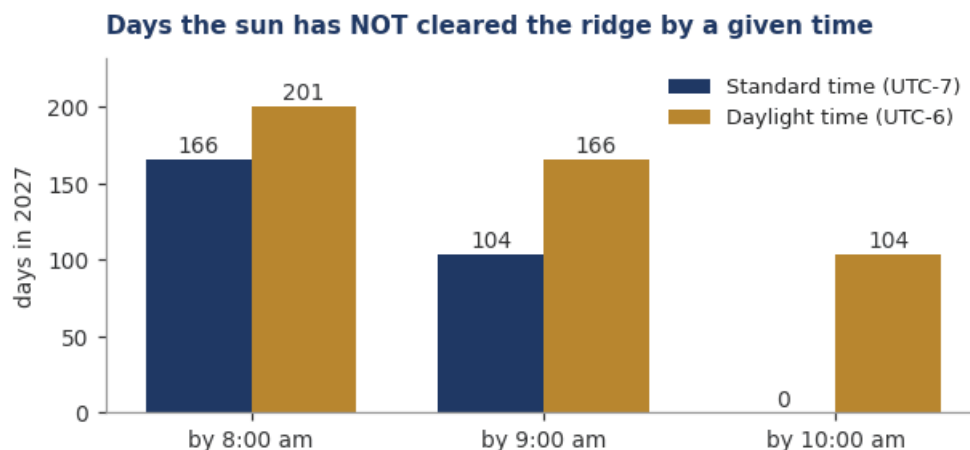


Figure 2. The same counts, drawn. Same total sunlight either way; a different distribution of it.

2.3 Through the year

Sun-on and sun-off times at the townsite with terrain included, for calendar year 2027. The winter solstice row is for 21 December 2026.

	Std on	Std off	hours	Dst on	Dst off	hours
Winter solstice	9:35 am	3:26 pm	5.87	10:35 am	4:26 pm	5.87
15 January	9:39 am	3:53 pm	6.25	10:39 am	4:53 pm	6.25
1 February	9:26 am	4:16 pm	6.85	10:26 am	5:16 pm	6.85
Spring equinox	7:49 am	5:52 pm	10.07	8:49 am	6:52 pm	10.07
Summer solstice	6:07 am	8:20 pm	14.23	7:07 am	9:20 pm	14.23
Fall equinox	7:25 am	5:38 pm	10.23	8:25 am	6:38 pm	10.23
15 November	9:05 am	3:40 pm	6.60	10:05 am	4:40 pm	6.60

2.4 The summer side of the ledger

Daylight time is not without advantage, and the same calculation measures it. In July and August the total hours of direct sun are identical under both options, but the evening end differs:

Days the sun is still on the town	Standard time	Daylight time
at 7 pm in July	31 of 31	31 of 31
at 8 pm in July	22 of 31	31 of 31
at 7 pm in August	17 of 31	31 of 31
at 8 pm in August	0 of 31	17 of 31

July averages 13.69 hours of direct sun a day and August 12.47, identical under both clocks. Usable light lasts roughly 40 to 50 minutes past the sun-off time in midsummer at this latitude, because twilight is long even once the sun is behind the ridge.

3. A separate question: light for walking to school

Whether there is enough light outdoors to see and be seen is not the same question as whether direct sunlight has reached the town, and the two must not be mixed. Ridges block direct sunlight, but they do not stop the sky lighting up. The figures in this section use civil dawn, the moment the sun's centre reaches 6 degrees below the horizon, on a flat horizon, which is the same convention published tables use.

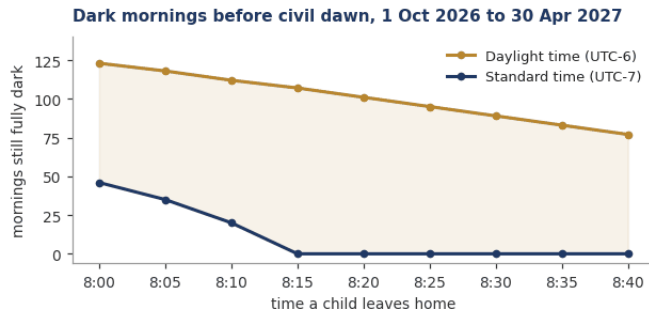
The counts below are for 1 October 2026 to 30 April 2027, a 212-day window, and give the number of mornings still fully dark at each school's own bell. Bell times were read from each school's posted schedule on 6 September 2026.

School	Bell	Standard time	Daylight time
Nicholson Elementary	8:40	0 mornings	77 mornings
classes begin	8:45	0	69
Golden Secondary	8:40	0	77
classes begin	8:45	0	69
Alexander Park	8:50	0	63
classes begin	8:55	0	55
Lady Grey Elementary	8:53	0	58
classes begin	8:58	0	49

A bell is an arrival time, so a count anchored to it measures whether it was dark when a child reached school, not whether it was dark while they walked. Nobody has measured when Golden children actually leave home, so rather than choose a departure time the whole plausible range is shown:

Leaving home at	Standard time	Daylight time	Difference
8:00 am	46	123	77
8:05 am	35	118	83
8:10 am	20	112	92
8:15 am	0	107	107
8:20 am	0	101	101
8:25 am	0	95	95
8:30 am	0	89	89
8:35 am	0	83	83
8:40 am	0	77	77

The absolute figure depends on an assumption nobody has measured. The difference between the two options does not: across the entire plausible range it stays between 77 and 107 mornings. Separately, the number of days on which sunrise itself falls after 8:45 am is 35 under standard time and 113 under daylight time.



Departure time is not measured, so the whole plausible range is shown. The gap between the two options stays between 77 and 107 mornings throughout.

Figure 5. The shaded band is the difference between the two options, and it never closes.

These civil dawn figures use a flat horizon and therefore carry the same blind spot as every published table. At civil dawn the twilight glow sits low in the eastern sky, which is precisely the band the eastern ridge occludes, so usable light on the ground in Golden arrives later than these figures show. The direction of that error is known; the magnitude is not computed here. The counts are therefore a floor under both options, not an estimate.

4. How the calculation works

The program does two separate jobs and then puts them together.

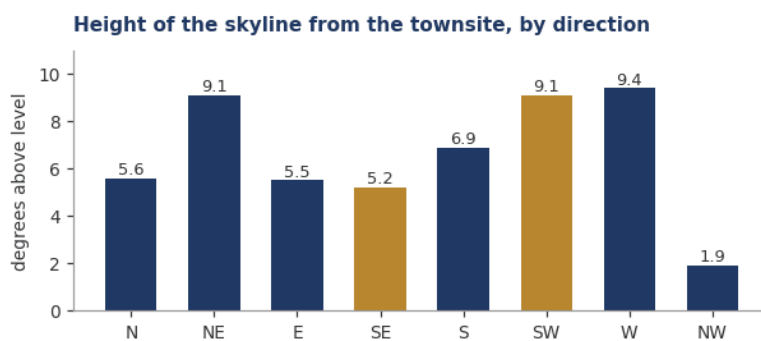
4.1 Where the sun is

Solar altitude and azimuth are computed with the NOAA Solar Calculator algorithm. This is pure arithmetic: it uses no data files and no external service, so it can be checked line by line against the published algorithm by anyone who wants to.

4.2 How high the mountains stand

For each of 720 compass bearings, half a degree apart, the program steps outward from the observation point in 40 metre increments to 60 kilometres, reading the ground elevation at each step. From each of those 1,500 sample points per bearing it subtracts the amount that point falls below a straight sightline because the Earth curves away and the atmosphere bends light, then computes the angle that point makes above the observer's horizontal. The largest angle along the whole ray is the height of the skyline in that direction.

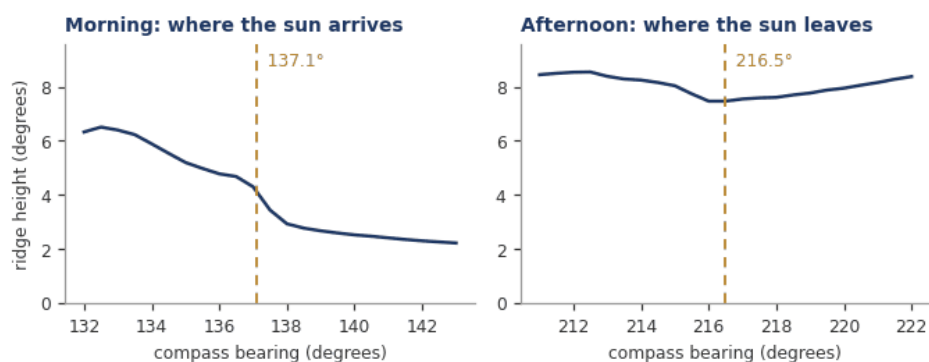
For the Golden townsite that produces the following skyline:



Eight-point summary only. Amber marks the quadrants the winter sun uses. Highest point overall is 12.2 degrees at bearing 240.

Figure 3. The valley in one picture: high walls west and northeast, the Columbia Valley opening to the northwest.

The highest point on the whole skyline is 12.2 degrees at bearing 240, to the southwest. That peak is nearly irrelevant to the winter solstice result, because the sun never gets there. On 21 December the sun runs from about 126 degrees at first light to about 231 degrees at last light, peaking at only 15.3 degrees altitude due south. The two bearings that actually decide the solstice times are 137.1 degrees, where the ridge stands at 4.50 degrees when the sun arrives, and 216.5 degrees, where it stands at 7.55 degrees when the sun leaves.



Measured skyline at half-degree steps, from skyline_full.csv. These two bands, not the 12.2 degree peak, decide the solstice times.

Figure 4. The measured skyline at the two bearings that decide the solstice times.

That asymmetry is why the afternoon loss of 73 minutes is larger than the morning loss of 42. It is a property of the shape of the valley, not of the calculation. It is also why the morning figure is the firmer of the two: the ridge falls away steeply where the sun arrives, so a small error in the skyline moves the time very little, while the afternoon skyline is nearly flat across ten degrees of bearing, so the same error moves the time considerably more.

4.3 Putting them together

The sun is counted as being on the town at any given moment when its altitude exceeds the height of the skyline at the compass bearing where it happens to be sitting. This is tested every minute across the full local day. The first minute the test passes is sun-on and the last is sun-off. The two clock options differ only by the UTC offset applied when converting local clock time, so the totals are necessarily identical between them.

5. Where the data came from

Elevation data is the AWS Terrain Tiles open dataset, accessed on 6 September 2026 from the skadi collection at s3.amazonaws.com/elevation-tiles-prod. It is a composite product originally assembled by Mapzen, SRTM-derived and void-filled at this latitude, at 1 arc-second resolution, which is roughly 30 metres per sample. It is not raw SRTMGL1, and anyone reproducing this should fetch from the same source rather than assume an unmodified SRTM tile will match.

Four tiles cover the area. Each is 3601 by 3601 samples, and the program trims the shared edges and mosaics them into a 7200 by 7200 grid. A SHA-256 fingerprint was recorded for each file so that a reader can confirm they are working from the same bytes; the first sixteen characters are given here and the record is in the repository.

Tile	Size (bytes)	SHA-256, first 16	Voids	Elevation range
N50W117	25,934,402	1b401a755c8f1e82	0	526 to 3405 m
N50W118	25,934,402	37743dd44cf11da5	0	426 to 3160 m
N51W117	25,934,402	6772decc7511f5c7	0	774 to 3531 m
N51W118	25,934,402	717bd1d1639e7dea	0	542 to 3488 m

Missing values, or voids, are a common defect in elevation data and would matter here, because the program skips them rather than filling them, so a void on a ridgeline would lower the computed skyline and understate the terrain effect. A scan of all 51,868,804 raw tile samples, of which 51,840,000 form the working mosaic after the shared edges are trimmed, found none in any of the four tiles.

The observation point is 51.2969 degrees north, 116.9647 degrees west. Its ground elevation is 788.0 metres, taken from the elevation data itself by interpolation rather than from a survey or a published figure, with 1.6 metres of eye height added, giving a sightline origin of 789.6 metres.

Attribution required by the dataset: SRTM data courtesy of the U.S. Geological Survey; contains information licensed under the Open Government Licence, Canada; and Mapzen for the Terrain Tiles collection itself.

Reference values used for checking come from the US Naval Observatory, Astronomical Applications Department.

6. How the work was checked

6.1 The solar arithmetic, against the US Naval Observatory

The program was run with the mountains switched off and its answers compared against the US Naval Observatory for the same coordinates and the same dates. On a flat horizon the answers must reproduce standard almanac times, so any disagreement would be a defect in the arithmetic rather than a fact about terrain.

Date	Event	This model	USNO	Difference
21 December 2026	sunrise	8:52:18	8:50	+2.32 min
	solar noon	12:46:05	12:46	+0.10 min
	sunset	4:39:53 pm	4:42 pm	-2.10 min
15 January 2027	sunrise	8:47:48	8:46	+1.82 min
20 March 2027	sunrise	6:52:12	6:51	+1.22 min
21 June 2027	sunrise	4:33:48	4:32	+1.82 min

The deviations are not merely small, they have the right shape. Every sunrise lands slightly late and every sunset slightly early, by a similar amount, with a spread of 1.10 minutes across all of them. That one-signed pattern is the expected signature of a documented difference in convention: the Naval Observatory marks sunrise when the upper edge of the sun's disc appears, while this program marks the moment the sun's centre clears the horizon, because the centre is the better test for direct light reaching the ground. Deviations scattered in sign or size would have indicated a real error. The test checks for that separately, and it also verifies that it can fail, by producing named failures when the tolerance is forced to an impossible value.

Solar noon is the strongest single check, because it involves no limb convention at all. It agrees with the Naval Observatory to within 0.6 minutes on all four dates.

6.2 The search distance

If the program stopped looking for terrain too close to town, a nearer and lower ridge would be mistaken for the skyline. The whole calculation was therefore run at four search distances:

Search radius	Solstice sun-on	Result
10 km	9:29 am	<i>too short, misses ridges</i>
20 km	9:35 am	<i>stable</i>
40 km	9:35 am	<i>identical to 20 km</i>
60 km	9:35 am	<i>identical to 20 km</i>

Every row of the output is identical from 20 kilometres outward. The reported figures use 60 kilometres. The answer is therefore not an artifact of where the search stopped.

6.3 Sampling fineness

Rays half a degree apart could in principle step over a narrow notch in a ridge, and the sample points along each ray are placed by a flat approximation rather than a true geodesic. Both were measured rather than argued about. Against rays ten times finer, the half-degree spacing differs by 0.023 degrees RMS on the morning arc and 0.018 on the afternoon. The flat approximation differs from a proper geodesic by at most 0.120 degrees over the arcs the sun actually uses.

Those numbers mean something once compared against how much a skyline error is worth. A quarter of a degree of error moves the solstice sunrise by about a minute. The measured errors are roughly a fifth of that. Even a full degree of error, forty times the measured RMS, moves the 9:35 am figure only to 9:37 or 9:29.

6.4 Reproducibility and independent review

Running the diagnostic twice regenerates its three output files byte for byte identically, which was confirmed on 6 September 2026. Separately, the method was put to an outside technical review, which raised six points. Two produced corrections to the published report, three were answered with measurements, and one was answered in full. Both the review response and the corrections are published in the repository, so a reader can see what was challenged and what changed rather than taking the outcome on trust.

7. What this does not establish

These should be stated alongside any figure quoted from this work.

- The ridge angles have not been checked against an independent measurement of the actual mountains. They have been shown to be stable, reproducible and internally consistent, and the solar half is validated against an outside authority, but no computed ridge angle has yet been compared with an independently known one. Closing that would need either surveyed horizon data or a timed observation of the sun clearing the ridge. This is the largest remaining gap.
- One observation point. Every figure describes the townsite coordinates given above. Times differ elsewhere in town and differ more on the benches.
- The elevation data is a surface model rather than bare earth, so it includes trees and buildings. On distant ridgelines that makes them slightly too tall and overstates the effect; at the observation point it raises the assumed viewpoint and understates it. Anything closer than about 30 metres does not appear at all. The biases pull in opposite directions and which dominates has not been established.
- Clear skies only. Valley cloud and December temperature inversions are not modelled, and in December they matter. Inversions also bend light more than the standard refraction coefficient allows, which would place true sunrise slightly earlier than calculated.
- The morning and afternoon figures are not equally firm. A quarter of a degree of skyline error is worth about a minute in the morning, because the ridge falls away steeply as the sun climbs, but roughly ten minutes in the afternoon, because the sun descends into a skyline that is rising again. Treat 9:35 am as firm and 3:26 pm as give or take a few minutes.
- Two of the four school bell schedules were read from pages headed 2025-26 and one states no year, so only Golden Secondary is on record as current. Morning bells rarely move between years, but that is an assumption rather than a check. The binding case, the earliest bell at 8:40, is the best-sourced of the four.
- The annual day counts use two-minute sampling, so individual days sitting close to a threshold could be off by one sample.

8. Reproducing this yourself

The code and the data record are public at github.com/LuminarAiConsultancy/golden-sunlight, under an MIT licence for the code and Creative Commons Attribution 4.0 for the documents. It needs Python and one common library, and no account, key or payment of any kind. The repository README carries a step by step walkthrough for a computer with nothing installed on it.

In short, from the repository folder:

Command	What it does
<code>pip install -r requirements.txt</code>	installs the one dependency
<code>python selftest.py</code>	checks the solar arithmetic against the US Naval Observatory

Command	What it does
python goldensun.py	the full model with terrain; downloads the elevation data once
python civildawn.py	the civil dawn and school bell figures
python season.py	month by month evening tables
python diagnostics.py	the provenance and sensitivity measurements in section 6

The first run of the third command downloads about 51 MB of elevation data and takes roughly 90 seconds. Everything after that runs in seconds. If any figure in this document cannot be reproduced from that repository, the discrepancy is worth raising, and I would want to know about it.

Joy Guyot, Councillor, Town of Golden | joy@luminaryx.ca | 6 September 2026

Personal work by a councillor. Not a Town of Golden publication. Provided as is, without warranty of any kind.